Project Title: Insurance Fraud Detection using ML

Project Design Phase-I - Solution Fit Template

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Define CS, fit into CC

1. CUSTOMER SEGMENT(S) CS

Who is your customer?
i.e. working parents of 0-5 y.o. kids

Insurance companies

Fraud investigation teams

Insurance data analysts

Claim verification departments

Insurance company managers

6. CUSTOMER CONSTRAINTS CC

What constraints prevent your customers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices.

Large number of insurance claims

Limited time for manual investigation

Lack of advanced analytical tools

Limited technical expertise in fraud detection systems

5. AVAILABLE SOLUTIONS AS

Which solutions are available to the customers when they face the problem
or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking

What solutions exist today:	Problems with these:
Manual fraud investigation by analysts	Slow
Rule-based fraud detection systems	Error-prone
Basic statistical analysis tools	Difficult to scale with large datasets
Traditional claim verification processes	

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

2. JOBS-TO-BE-DONE / PROBLEMS J&P

Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides.

Detect fraudulent insurance claims

Reduce financial losses caused by fraud

Analyze large volumes of insurance claim data

Identify suspicious patterns in claim submissions

Improve accuracy of claim verification

9. PROBLEM ROOT CAUSE RC

What is the real reason that this problem exists?
What is the back story behind the need to do this job?
i.e. customers have to do it because of the change in regulations.

Large volume of insurance claims

Fraudsters exploiting weaknesses in claim systems

Lack of automated fraud detection methods

Manual verification processes being slow and inefficient

7. BEHAVIOUR BE

What does your customer do to address the problem and get the job done?
i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace)

Review claim documents manually

Investigate suspicious claims

Verify accident reports and supporting documents

Use existing internal software for claim management

Focus on J&P, tap into BE, understand RC

Identify strong TR & EM

3. TRIGGERS TR

What triggers customers to act? i.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news.

Financial losses due to fraud

Manual investigation taking too much time

4. EMOTIONS: BEFORE / AFTER EM

How do customers feel when they face a problem or a job and afterwards?
i.e. lost, insecure > confident, in control - use it in your communication strategy & design.

Before	After
Frustrated	Confident in fraud detection
Overwhelmed by large datasets	Efficient claim analysis
Concerned about fraud losses	Reduced workload
Stressed due to manual investigation	Better financial protection

10. YOUR SOLUTION SL

If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality.
If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.

Key features:

Automated fraud prediction using ML models

Web-based interface for entering claim details

Real-time fraud prediction results

Data-driven analysis of claim patterns

Faster and more accurate fraud detection

Benefits:

Reduces investigation time

Improves fraud detection accuracy

Helps insurance companies minimize financial losses

8. CHANNELS of BEHAVIOUR CH

8.1 ONLINE
What kind of actions do customers take online? Extract online channels from #7

8.2 OFFLINE
What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.

ONLINE:	OFFLINE:
Insurance claim management systems	Manual document verification
Company databases	Investigator interviews
Fraud detection software tools	Reviewing physical accident reports
Internal digital dashboards	Internal meetings and audits

Identify strong TR & EM